

A LENGTH-14 NEGATIVE ANSWER TO ALDERSON'S $q = 4$ OPTIMALITY QUESTION

ABSTRACT. Alderson constructed an extendable, additively maximal additive $(112, 2, 104)_{16/4}$ -code and asked whether length 112 is optimal for $q = 4$. A classical spread of $\text{PG}(3, 4) \setminus \text{PG}(3, 2)$ yields a faithful additive MDS $(14, 2, 13)_{16/4}$ -code that is extendable as a 16-ary code but admits no $\mathbb{F}_4$ -additive extension. Hence the answer is negative and the minimum length in the $q = 4$ clause of Alderson's Problem 9.2 is at most 14. Problem 9.2 itself notes that any sub-multiset of its 112 external lines whose maximal-fold points still meet every line yields a shorter example; that condition governs additive maximality alone, and what is supplied here is a sub-multiset for which extendability, the other requirement, also holds. We do not determine the global minimum.

1. RESULT

Alderson's Theorem 6.6 gives, for $t = 2$, an extendable, additively maximal additive $(112, 2, 104)_{16/4}$ -code, whose 112 coordinates are indexed by the lines of $\Pi = \text{PG}(3, 4)$ external to a scattered $\mathbb{F}_2$ -linear set. Problem 9.2 of [1] then reads, in full:

Extend Theorem 6.6 to non-square, non-prime q . Further, determine the minimum length of an extendable, additively maximal additive $(n, 2, d)_{q^2/q}$ -code for square q : is $n = 112$ optimal for $q = 4$? (Any sub-multiset of the external lines whose set of maximal-fold points still meets every line of Π yields a shorter example.)

Theorem 1 below answers the final clause negatively, at length 14; the first clause and the minimum for general square q are untouched. Remark 3 records what the closing parenthetical already supplies and what it does not.

The geometric input is classical: van Dam [3] classified the spreads of $\text{PG}(3, 4) \setminus \text{PG}(3, 2)$ into three orbits under the collineation group $\text{P}\Gamma\text{L}(4, 4)$, and Mellinger [4] later obtained the corresponding three mixed partitions consisting of one Baer subgeometry and fourteen lines.

Theorem 1. *There exists a faithful additive MDS $(14, 2, 13)_{16/4}$ -code C that is extendable as a 16-ary code but is additively maximal. Its Hamming weight enumerator is*

$$W_C(z) = 1 + 210z^{13} + 45z^{14}.$$

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Proof. Let $\mathbb{F}_2 \subset \mathbb{F}_4 \subset \mathbb{F}_{16}$, choose $\omega \in \mathbb{F}_4$ with $\omega^2 + \omega + 1 = 0$, put $E = \mathbb{F}_4^4$, and let $U = \mathbb{F}_2^4 \leq E$. In $\Pi = \text{PG}(E) = \text{PG}(3, 4)$, define the standard Baer subgeometry

$$B = \{\langle u \rangle_{\mathbb{F}_4} : 0 \neq u \in U\} \cong \text{PG}(3, 2).$$

Thus $|B| = 15$. The $\mathbb{F}_4$ -line through two distinct points of B meets B in the three points of the corresponding $\mathbb{F}_2$ -subline. In particular, B contains no $\mathbb{F}_4$ -line.

We first give an explicit representative of the classical complement spreads classified in [3, 4]. Let

$$T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \text{GL}(4, 2) \leq \text{GL}(4, 4).$$

The upper-left block has minimal polynomial $X^3 + X + 1$, so T has order 7. Put

$$\begin{aligned} u &= (1, 0, 0, \omega)^\top, & v &= (0, 1, \omega^2, 0)^\top, & L &= \langle u, v \rangle_{\mathbb{F}_4}, \\ u' &= (1, 0, 0, \omega^2)^\top, & v' &= (0, 1, \omega, 0)^\top, & L' &= \langle u', v' \rangle_{\mathbb{F}_4}. \end{aligned}$$

The five points of $\text{PG}(L)$ are represented by $u, v, u + v, u + \omega v, u + \omega^2 v$. These representatives are normalized at their first nonzero coordinate, and each has a coordinate outside $\mathbb{F}_2$; hence none lies in B . The same holds for $\text{PG}(L')$, its Frobenius conjugate. Since T has entries in $\mathbb{F}_2$, every line $\text{PG}(T^j L)$ and $\text{PG}(T^j L')$ is disjoint from B .

For vectors $x_1, \dots, x_4$, write $\det[x_1, x_2, x_3, x_4]$ for the determinant of the matrix with those columns. A direct calculation gives

$$\begin{aligned} (\det[u, v, T^k u, T^k v])_{k=1}^6 &= (\omega, 1, \omega^2, \omega^2, 1, \omega), \\ (\det[u', v', T^k u', T^k v'])_{k=1}^6 &= (\omega^2, 1, \omega, \omega, 1, \omega^2), \\ (\det[u, v, T^k u', T^k v'])_{k=0}^6 &= (1, \omega, \omega, \omega^2, \omega, \omega^2, \omega^2). \end{aligned}$$

All entries are nonzero. Since intersections depend only on relative powers of T , these three calculations show that the fourteen lines

$$\ell_{j+1} = \text{PG}(T^j L), \quad \ell_{j+8} = \text{PG}(T^j L') \quad (0 \leq j \leq 6)$$

are pairwise disjoint. They contain $14 \cdot 5 = 70 = 85 - 15$ points, all outside B , and therefore

$$(1) \quad \Pi = B \dot{\cup} \ell_1 \dot{\cup} \dots \dot{\cup} \ell_{14}.$$

Write $\ell_i = \text{PG}(V_i)$, where $V_i \leq E$ has $\mathbb{F}_4$ -dimension 2. Identifying $\mathbb{F}_{16}$ with a two-dimensional $\mathbb{F}_4$ -space, choose an $\mathbb{F}_4$ -linear epimorphism

$$\rho_i : E \longrightarrow \mathbb{F}_{16}, \quad \ker \rho_i = V_i,$$

and define

$$c(a) = (\rho_1(a), \dots, \rho_{14}(a)), \quad C = \{c(a) : a \in E\}.$$

Since the lines are disjoint,

$$\ker c = \bigcap_{i=1}^{14} V_i \subseteq V_1 \cap V_2 = \{0\}.$$

Hence c is injective and $|C| = 4^4 = 16^2$. Each coordinate map is surjective, so C is faithful.

For $a \neq 0$, the zero coordinates of $c(a)$ are exactly the indices i for which $\langle a \rangle_{\mathbb{F}_4} \in \ell_i$. By (1), $c(a)$ has weight 14 when $\langle a \rangle_{\mathbb{F}_4} \in B$ and weight 13 otherwise. Since $|\Pi| = 85$, there are $3|B| = 45$ nonzero words of weight 14 and $3(85 - 15) = 210$ of weight 13. This proves the stated weight enumerator and minimum distance. Since $13 = 14 - 2 + 1$, the code is MDS.

Viewed over $\mathbb{F}_2$, both E/U and $\mathbb{F}_{16}$ have dimension 4. Choose an $\mathbb{F}_2$ -linear isomorphism $\tau : E/U \rightarrow \mathbb{F}_{16}$ and set, following the coset argument in the proof of [1, Theorem 6.6],

$$C^+ = \{(c(a), \tau(a + U)) : a \in E\} \subseteq \mathbb{F}_{16}^{15}.$$

For distinct $a, b \in E$, put $x = a - b$. If $x \in U$, then the new symbols agree and $\langle x \rangle_{\mathbb{F}_4} \in B$, so $d_H(c(a), c(b)) = 14$. If $x \notin U$, then the new symbols differ and $d_H(c(a), c(b)) \geq 13$. Thus $d(C^+) = 14$, with equality in the first case, and puncturing the last coordinate gives C . Hence C is extendable; in fact, this extension is $\mathbb{F}_2$ -linear, the subfield phenomenon of [1, Remark 6.7]. So C , regarded not as an $\mathbb{F}_4$ -linear but as an $\mathbb{F}_2$ -linear subspace of $\mathbb{F}_{16}^{14}$ — that is, as an additive $(14, 2, 13)_{16/2}$ -code in the general $(n, k, d)_{q^m/q}$ notation of [1], taken there with $q = 2$ and $m = 4$, so that the $\mathbb{F}_2$ -dimension is $km = 8$, as it is here since $|C| = 2^8$ — does admit an additive extension: the additive maximality proved next is relative to the declared field of linearity $\mathbb{F}_4$.

Suppose that C admitted an $\mathbb{F}_4$ -additive extension $\tilde{C} \leq \mathbb{F}_{16}^{15}$ of minimum distance 14. Puncturing the new coordinate is an $\mathbb{F}_4$ -linear bijection $\tilde{C} \rightarrow C$. Therefore, for some $\mathbb{F}_4$ -linear map $\rho : E \rightarrow \mathbb{F}_{16}$,

$$\tilde{C} = \{(c(a), \rho(a)) : a \in E\}.$$

If $\langle a \rangle_{\mathbb{F}_4} \notin B$, then $\text{wt}(c(a)) = 13$, so $\rho(a) \neq 0$. Consequently every nonzero vector in $\ker \rho$ has projective direction in B . But $\dim_{\mathbb{F}_4} \mathbb{F}_{16} = 2$, and hence

$$\dim_{\mathbb{F}_4} \ker \rho \geq 4 - 2 = 2.$$

The projectivization of a two-dimensional $\mathbb{F}_4$ -subspace of $\ker \rho$ is an $\mathbb{F}_4$ -line contained in B , a contradiction. Thus C has no $\mathbb{F}_4$ -additive extension. $\square$

Corollary 2. *The minimum length asked for in the $q = 4$ clause of Alderson's Problem 9.2 is at most 14. In particular, 112 is not optimal.*

Proof. Theorem 1 has exactly the required extension and additive maximality properties. $\square$

Remark 3. Problem 9.2 already states that any sub-multiset of the 112 external lines whose set of maximal-fold points still meets every line of Π yields a shorter example, and the code of Theorem 1 is such a sub-multiset:

each of $\ell_1, \dots, \ell_{14}$ is taken once, so by (1) the maximal fold is 1 and the maximal-fold set is $\Pi \setminus B$, which meets every line of Π because B contains no $\mathbb{F}_4$ -line. For $t = 2$ Alderson's scattered $\mathbb{F}_2$ -linear set is the projectivization of $\{(x, y, x^2, y^2) : x, y \in \mathbb{F}_4\}$, an $\mathbb{F}_2$ -subspace of E whose $\mathbb{F}_2$ -basis is also an $\mathbb{F}_4$ -basis of E , hence the image of $U = \mathbb{F}_2^4$ under an element of $\text{GL}(4, 4)$; it is therefore a Baer subgeometry $\text{PG}(3, 2)$ of Π (cf. [1, Lemma 6.4]), projectively equivalent to the set B above. The lines of Π external to a Baer subgeometry number exactly 112: of the 357 lines of Π , 112 miss B , 210 meet it in one point and 35 in three. That count is Alderson's $n_2 = 112$, so $\ell_1, \dots, \ell_{14}$ are fourteen of his own 112 external lines, each taken once; the maximal-fold set $\Pi \setminus B$ is his; and C^+ is his coset construction.

What that parenthetical supplies is one of the two requirements. Its condition, that the maximal-fold points meet every line of Π , is the criterion for additive maximality of [1, Corollary 3.6], in which the flats to be met are the $(km - m - 1)$ -flats of Π and hence, for $k = m = 2$, its lines; it is silent on extendability, which Problem 9.2 also demands, so a sub-multiset meeting the condition need not answer the problem. The content of Theorem 1 is accordingly the choice of sub-multiset: a line-Baer partition of Π makes the fourteen chosen lines pairwise disjoint, so the code is MDS, and leaves the uncovered points equal to the Baer subgeometry B , and it is that hole set which makes the 16 cosets of U available as the fibres of a 16-ary extension of minimum distance 14.

Remark 4. The corollary answers only the final yes-or-no clause of Alderson's Problem 9.2, refuting the optimality of 112 for $q = 4$. It does not determine the global minimum or address the nonsquare, nonprime case, and no lower bound is proved here. The spread in (1) is classical; the contribution is its coding-theoretic consequence in Alderson's extension framework. In the projective-systems dictionary used in [1], a faithful additive MDS $(n, 2, n - 1)_{q^2/q}$ -code is a partial spread of n lines of $\text{PG}(3, q)$, and additive maximality corresponds, by [1, Corollary 3.6] again, to that partial spread being maximal; the object here is thus a maximal partial spread of $\text{PG}(3, 4)$ of size 14 whose holes form a Baer subgeometry. Additive MDS codes over $\mathbb{F}_{16}$ that are $\mathbb{F}_4$ -linear are studied in [2], where the longest ones are classified; additive extendability in the sense of [1] is not considered there.

EXACT VERIFICATION

Everything needed to redo the computation is printed above: the relation $\omega^2 + \omega + 1 = 0$, the matrix T , the vectors u, v, u', v' , and the partition (1). From these one forms the fourteen lines $\ell_{j+1} = \text{PG}(T^j L)$ and $\ell_{j+8} = \text{PG}(T^j L')$, checks the nineteen determinants displayed in the proof, tests all 91 pairs of lines for disjointness, builds c , C and C^+ as in the proof, recovers $W_C(z) = 1 + 210z^{13} + 45z^{14}$ and the minimum distances $d(C) = 13$ and $d(C^+) = 14$, and finally runs over the $\mathbb{F}_4$ -linear maps $\rho : E \rightarrow \mathbb{F}_{16}$ to

confirm that none of them extends C to minimum distance 14. A referee can therefore reproduce every number printed here from this paper alone.

The exhibited object and every count printed above were recomputed on that basis independently, on a separate machine, by a program written from the definitions of this note; all 71 of its checks agree with the paper. The enumerations are exhaustive rather than sampled: all 85 points and all 357 lines of Π , all 91 pairs of the fourteen lines, all 256 words of C and of C^+ with all 32,640 unordered pairs for each minimum distance, all 357 two-dimensional $\mathbb{F}_4$ -subspaces of E , and, in the maximality step, all $4^8 = 65,536$ $\mathbb{F}_4$ -linear maps $E \rightarrow \mathbb{F}_{16}$, the completeness of that last list being itself checked. The run repeats the construction for other choices of $\rho_1, \dots, \rho_{14}$ and of τ , reports failures on seven deliberately corrupted variants of the object, and passes a positive control in which the hole set does contain a line: replacing $\ell_1, \dots, \ell_{14}$ by fourteen of the seventeen lines of a spread of $\text{PG}(3, 4)$ leaves as holes the three remaining, pairwise disjoint lines, and the same exhaustive search then accepts exactly $3|\text{GL}(2, 4)| = 540$ of the 65,536 maps ρ , namely, for each hole line $\text{PG}(V)$, the 180 maps with kernel V . So the negative answer for the Baer hole set is not an artifact of the search. The program and its transcript accompany this note; nothing above depends on them, since every quantity quoted is either printed here or recomputable from the data just listed.

What the re-run does not do. It does not re-derive anything from [1]: Theorem 6.6, the length 112, and the definitions of extendable and of additively maximal are taken from that paper; and the identification in Remark 3 of Alderson's scattered $\mathbb{F}_2$ -linear set with a Baer subgeometry projectively equivalent to B is argued there rather than checked by the program, which counts the 112 lines external to B itself. It does not verify the classifications of [3, 4], for which the explicit spread above is a self-contained substitute. And faithfulness is tested only in the sense used here, that every coordinate projection is onto $\mathbb{F}_{16}$. No search for a lower bound or for the global minimum was attempted. The re-run and this paper agree; that is agreement, not a proof.

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